In [1]:

```
import pandas as pd

data = {
     "Month": ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
     "Sales":  [10000, 12000, 15000, 13000, 17000, 16000],
     "profit": [2000, 3000, 4000, 2500, 3500, 3000]
}

df = pd.DataFrame(data)
print(df)
```

```
  Month  Sales  profit
0   Jan  10000    2000
1   Feb  12000    3000
2   Mar  15000    4000
3   Apr  13000    2500
4   May  17000    3500
5   Jun  16000    3000
```

In [2]:

```
df[['Month', 'Sales']]
```

Out[2]:

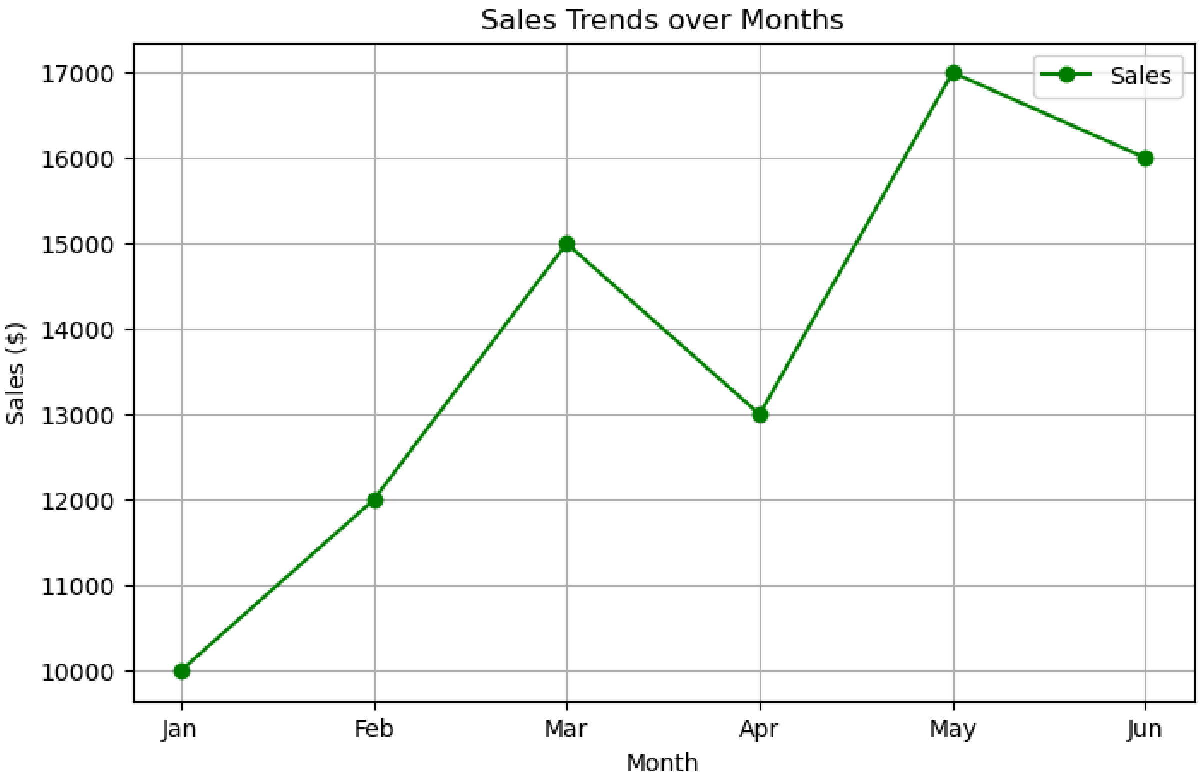
	Month	Sales
0	Jan	10000
1	Feb	12000
2	Mar	15000
3	Apr	13000
4	May	17000
5	Jun	16000

1. LINE PLOT SALES OVER TIME

In [3]:

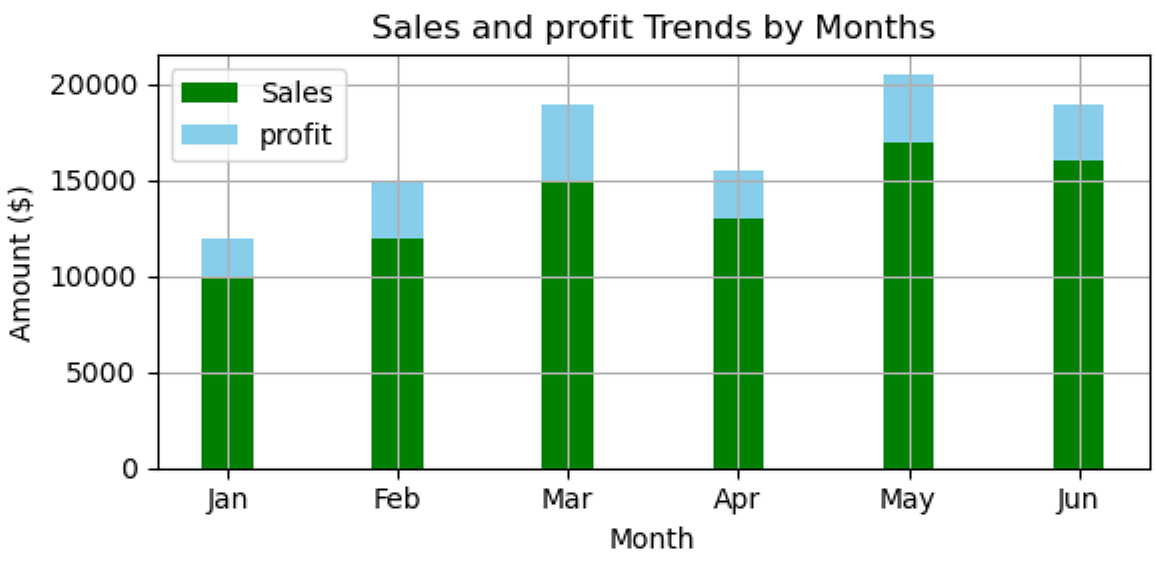
```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.plot(df['Month'], df['Sales'], color='Green', marker='o', linestyle='-', label=
plt.title('Sales Trends over Months')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid(True)
plt.legend()
plt.show()
```



2. BARPLOT MONTH VS PROFIT

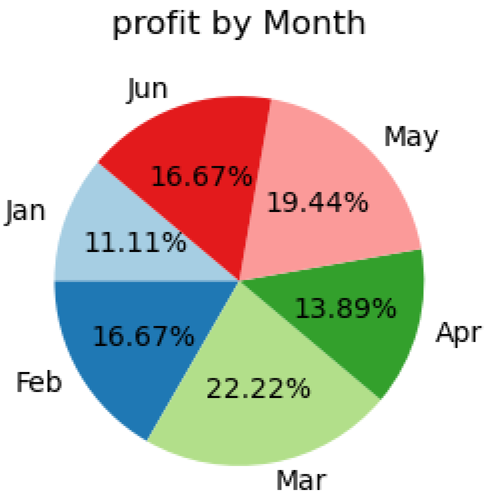
In [7]:
```
plt.figure(figsize=(6,3))
width = 0.3
plt.bar(df['Month'], df['Sales'], width=width, color='Green', label='Sales')
plt.bar(df['Month'], df['profit'], width=width, color='skyblue', label='profit', bo
plt.title('Sales and profit Trends by Months')
plt.xlabel('Month')
plt.ylabel('Amount ($)')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```



3. PIE CHART PROFIT VS MONTH

In [15]:

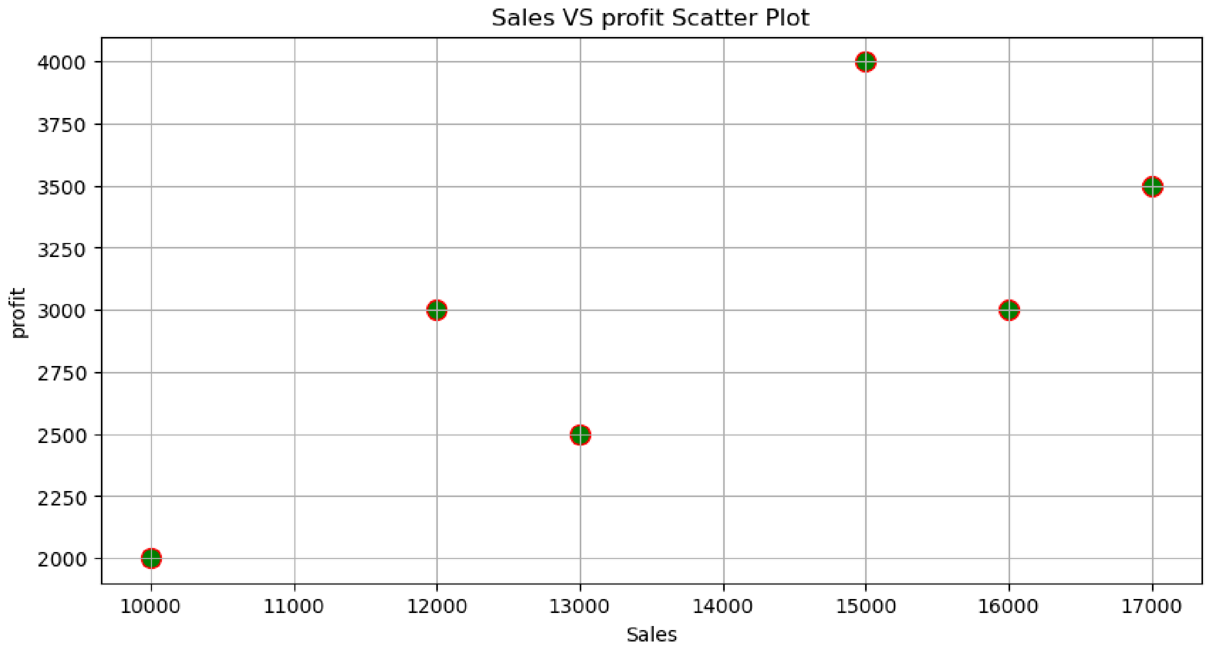
```
from enum import auto
plt.figure(figsize=(6,3))
plt.pie(df['profit'], labels=df['Month'], autopct ='%1.2f%%', startangle = 140, col
plt.title('profit by Month')
plt.show()
```



4. SCATTER PLOT

In [18]:

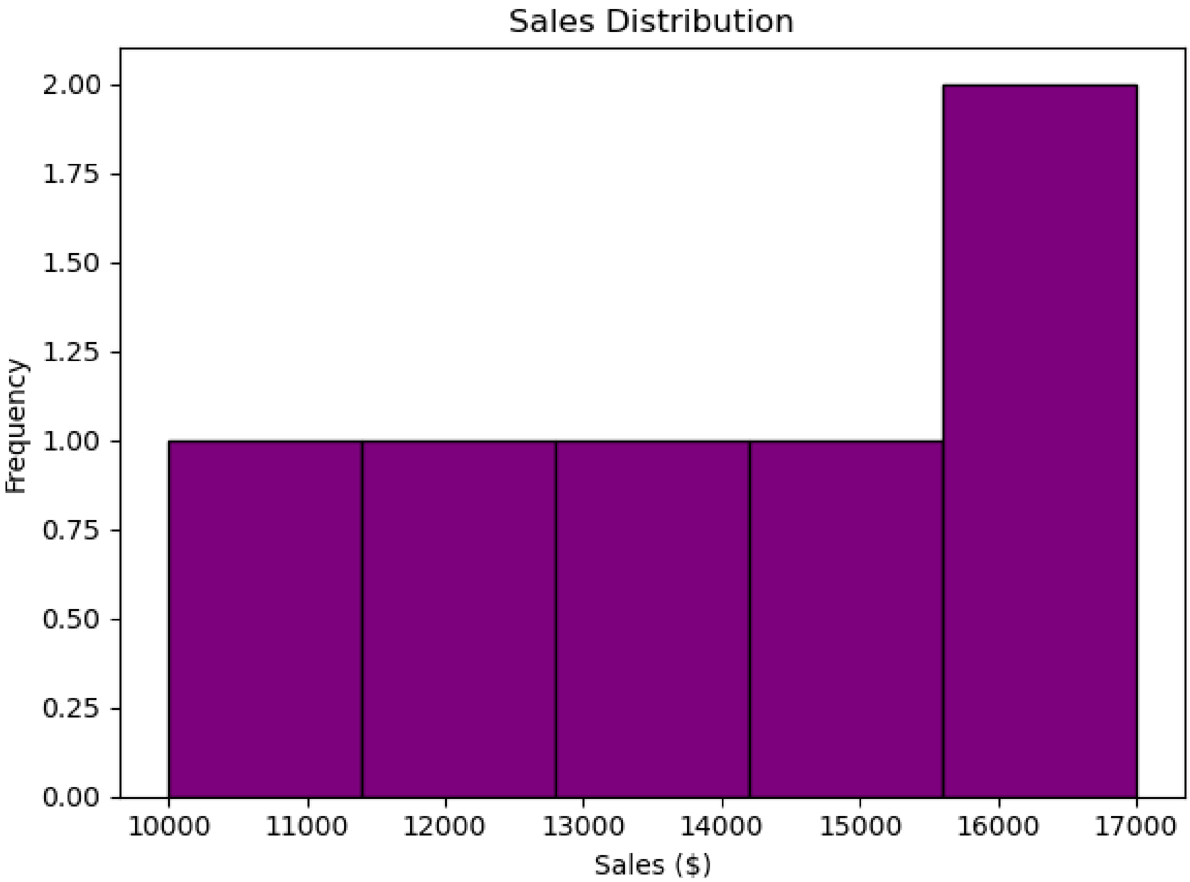
```
plt.figure(figsize=(10,5))
plt.scatter(df['Sales'], df['profit'], color='green', s=100, edgecolors='red')
plt.title('Sales VS profit Scatter Plot')
plt.xlabel('Sales')
plt.ylabel('profit')
plt.grid(True)
plt.show()
```



5. HISTOGRAM : DISTRIBUTION OF SALES

In [26]:

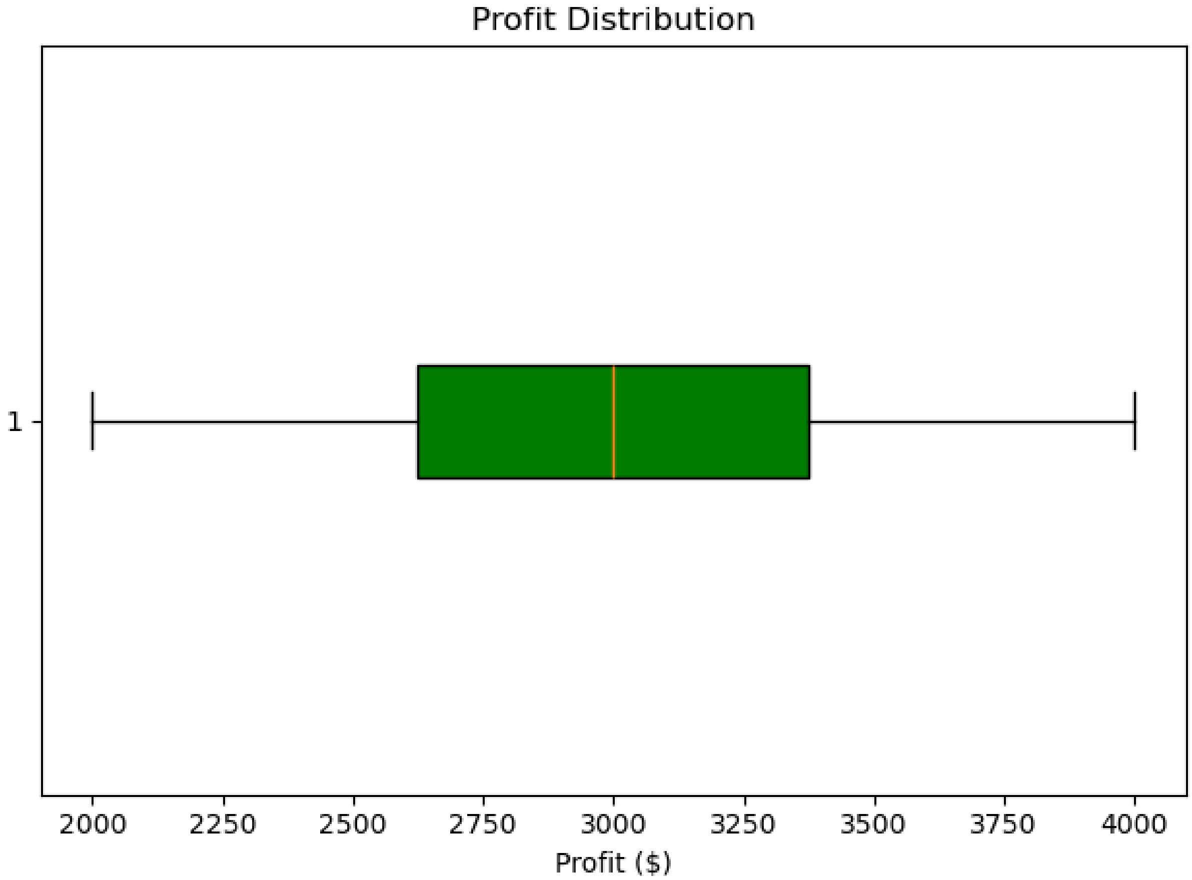
```
plt.Figure(figsize=(2,3))
plt.hist(df['Sales'], bins=5, color='purple', edgecolor='black')
plt.title('Sales Distribution')
plt.xlabel('Sales ($)')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



6. BOX PLOT : PROFIT DISTRIBUTION

In [27]:

```
plt.Figure(figsize=(3,2))
plt.boxplot(df['profit'], vert=False, patch_artist=True, boxprops =dict(facecolor="
plt.title('Profit Distribution')
plt.xlabel('Profit ($)')
plt.tight_layout()
plt.show()
```



In [28]:

```
pip install gradio
```

```
Requirement already satisfied: gradio in d:\users\3star\lib\site-packages (6.11.0)
Requirement already satisfied: aiofiles<25.0,>=22.0 in d:\users\3star\lib\site-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in d:\users\3star\lib\site-packages (from gradio) (4.10.0)
Requirement already satisfied: audioop-lts<1.0 in d:\users\3star\lib\site-packages (from gradio) (0.2.2)
Requirement already satisfied: brotli>=1.1.0 in d:\users\3star\lib\site-packages (from gradio) (1.2.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in d:\users\3star\lib\site-packages (from gradio) (0.135.3)
Requirement already satisfied: ffmpy in d:\users\3star\lib\site-packages (from gradio) (1.0.0)
Requirement already satisfied: gradio-client==2.4.0 in d:\users\3star\lib\site-packages (from gradio) (2.4.0)
Requirement already satisfied: groovy~=0.1 in d:\users\3star\lib\site-packages (from gradio) (0.1.2)
Requirement already satisfied: hf-gradio<1.0,>=0.3.0 in d:\users\3star\lib\site-packages (from gradio) (0.3.0)
Requirement already satisfied: httpx<1.0,>=0.24.1 in d:\users\3star\lib\site-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=0.33.5 in d:\users\3star\lib\site-packages (from gradio) (1.10.1)
Requirement already satisfied: jinja2<4.0 in d:\users\3star\lib\site-packages (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0,>=2.0 in d:\users\3star\lib\site-packages (from gradio) (3.0.2)
Requirement already satisfied: numpy<3.0,>=1.0 in d:\users\3star\lib\site-packages (from gradio) (2.3.5)
Requirement already satisfied: orjson~=3.0 in d:\users\3star\lib\site-packages (from gradio) (3.11.7)
Requirement already satisfied: packaging in d:\users\3star\lib\site-packages (from gradio) (25.0)
Requirement already satisfied: pandas<4.0,>=1.0 in d:\users\3star\lib\site-packages (from gradio) (2.3.3)
Requirement already satisfied: pillow<13.0,>=8.0 in d:\users\3star\lib\site-packages (from gradio) (12.0.0)
Requirement already satisfied: pydantic<=3.0,>=2.0 in d:\users\3star\lib\site-packages (from gradio) (2.12.4)
Requirement already satisfied: pydub in d:\users\3star\lib\site-packages (from gradio) (0.25.1)
Requirement already satisfied: python-multipart>=0.0.18 in d:\users\3star\lib\site-packages (from gradio) (0.0.24)
Requirement already satisfied: pytz>=2017.2 in d:\users\3star\lib\site-packages (from gradio) (2025.2)
Requirement already satisfied: pyyaml<7.0,>=5.0 in d:\users\3star\lib\site-packages (from gradio) (6.0.3)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.7 in d:\users\3star\lib\site-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in d:\users\3star\lib\site-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<2.0,>=0.40.0 in d:\users\3star\lib\site-packages (from gradio) (1.0.0)
Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in d:\users\3star\lib\site-packages (from gradio) (0.13.3)
Requirement already satisfied: typer<1.0,>=0.12 in d:\users\3star\lib\site-packages
```

```
(from gradio) (0.20.0)
Requirement already satisfied: typing-extensions~=4.0 in d:\users\3star\lib\site-packages (from gradio) (4.15.0)
Requirement already satisfied: uvicorn>=0.14.0 in d:\users\3star\lib\site-packages
(from gradio) (0.44.0)
Requirement already satisfied: fsspec in d:\users\3star\lib\site-packages (from gradio-client==2.4.0->gradio) (2025.10.0)
Requirement already satisfied: idna>=2.8 in d:\users\3star\lib\site-packages (from anyio<5.0,>=3.0->gradio) (3.11)
Requirement already satisfied: sniffio>=1.1 in d:\users\3star\lib\site-packages (from anyio<5.0,>=3.0->gradio) (1.3.0)
Requirement already satisfied: typing-inspection>=0.4.2 in d:\users\3star\lib\site-packages (from fastapi<1.0,>=0.115.2->gradio) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in d:\users\3star\lib\site-packages (from fastapi<1.0,>=0.115.2->gradio) (0.0.4)
Requirement already satisfied: certifi in d:\users\3star\lib\site-packages (from httpx<1.0,>=0.24.1->gradio) (2026.1.4)
Requirement already satisfied: httpcore==1.* in d:\users\3star\lib\site-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.9)
Requirement already satisfied: h11>=0.16 in d:\users\3star\lib\site-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.16.0)
Requirement already satisfied: filelock>=3.10.0 in d:\users\3star\lib\site-packages
(from huggingface-hub<2.0,>=0.33.5->gradio) (3.20.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in d:\users\3star\lib\site-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (1.4.3)
Requirement already satisfied: tqdm>=4.42.1 in d:\users\3star\lib\site-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (4.67.1)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\users\3star\lib\site-packages (from pandas<4.0,>=1.0->gradio) (2.9.0.post0)
Requirement already satisfied: tzdata>=2022.7 in d:\users\3star\lib\site-packages (from pandas<4.0,>=1.0->gradio) (2025.2)
Requirement already satisfied: annotated-types>=0.6.0 in d:\users\3star\lib\site-packages (from pydantic<=3.0,>=2.0->gradio) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.5 in d:\users\3star\lib\site-packages (from pydantic<=3.0,>=2.0->gradio) (2.41.5)
Requirement already satisfied: click>=8.0.0 in d:\users\3star\lib\site-packages (from typer<1.0,>=0.12->gradio) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in d:\users\3star\lib\site-packages (from typer<1.0,>=0.12->gradio) (1.5.4)
Requirement already satisfied: rich>=10.11.0 in d:\users\3star\lib\site-packages (from typer<1.0,>=0.12->gradio) (14.2.0)
Requirement already satisfied: colorama in d:\users\3star\lib\site-packages (from click>=8.0.0->typer<1.0,>=0.12->gradio) (0.4.6)
Requirement already satisfied: six>=1.5 in d:\users\3star\lib\site-packages (from python-dateutil>=2.8.2->pandas<4.0,>=1.0->gradio) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in d:\users\3star\lib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in d:\users\3star\lib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in d:\users\3star\lib\site-packages (from
markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0,>=0.12->gradio) (0.1.2)
Note: you may need to restart the kernel to use updated packages.
```

In [41]:

```python
import gradio as gr
import pandas as pd
import matplotlib.pyplot as plt
```

```
data = {
    "Month": ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    "Sales": [10000, 12000, 15000, 13000, 17000, 16000],
    "profit": [2000, 3000, 4000, 2500, 3500, 3000]
}

df = pd.DataFrame(data)

def generate_plot(plot_type):
    fig = plt.figure(figsize=(8,5))

    if plot_type == 'Line plot':
        plt.plot(df['Month'], df['Sales'], color='green', marker='o', linestyle='-'
        plt.title('Sales Trends over Months')
        plt.xlabel('Month')
        plt.ylabel('Sales ($)')
        plt.grid(True)
        plt.legend()

    elif plot_type == 'stacked bar Chart':
        fig.set_size_inches(10,6)
        width = 0.3
        plt.bar(df['Month'], df['Sales'], width=width, color='green', label='Sales'
        plt.bar(df['Month'], df['profit'], width=width, color='skyblue', label='Pro
        plt.title('Sales and Profit Trends by Months')
        plt.xlabel('Month')
        plt.ylabel('Amount ($)')
        plt.grid(True)
        plt.legend()
        plt.tight_layout()

    elif plot_type == 'Pie chart':
        fig.set_size_inches(7,7)
        plt.pie(df['profit'], labels=df['Month'], autopct='%1.2f%%',
                startangle=140, colors=plt.cm.Paired.colors)
        plt.title('Profit by Month')

    elif plot_type == 'Scatter Plot':
        plt.scatter(df['Sales'], df['profit'], color='green', s=100, edgecolors='re
        plt.title('Sales vs Profit Scatter Plot')
        plt.xlabel('Sales')
        plt.ylabel('Profit')
        plt.grid(True)

    elif plot_type == 'Histogram':
        plt.hist(df['Sales'], bins=5, color='purple', edgecolor='black')
        plt.title('Sales Distribution')
        plt.xlabel('Sales ($)')
        plt.ylabel('Frequency')

    elif plot_type == 'Box plot':
        plt.boxplot(df['profit'], vert=False, patch_artist=True,
                    boxprops=dict(facecolor="green"))
        plt.title('Profit Distribution')
        plt.xlabel('Profit ($)')
```

```
    plt.tight_layout()
    return fig


# ✅ Gradio UI (OUTSIDE function)
demo = gr.Interface(
    fn=generate_plot,
    inputs=gr.Radio(['Line plot', 'stacked bar Chart', 'Pie chart', 'Scatter Plot',
    outputs=gr.Plot(label="Sales Data Visualization"),
    title='Sales & Profit Visual Insight',
    description='Choose the type to visualize the data'
)

demo.launch()
```

```
* Running on local URL:  http://127.0.0.1:7860
* To create a public link, set `share=True` in `launch()`.
```

Out[41]:

In [40]:

```
import gradio as gr
import pandas as pd
import matplotlib.pyplot as plt

data = {
```

```
     "Month": ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
     "Sales":  [10000, 12000, 15000, 13000, 17000, 16000],
     "profit": [2000, 3000, 4000, 2500, 3500, 3000]
}

df = pd.DataFrame(data)

def generate_plot(plot_type):
    fig = plt.figure(figsize=(8,5))

    if plot_type == 'Line plot':
       plt.plot(df['Month'], df['Sales'], color='Green', marker='o', linestyle='-',
       plt.title('Sales Trends over Months')
       plt.xlabel('Month')
       plt.ylabel('Sales ($)')
       plt.grid(True)
       plt.legend(),

    elif plot_type == 'stacked bar Chart':
        fig.set_size_inches(10,6)
        width = 0.3
        plt.bar(df['Month'], df['Sales'], width=width, color='Green', label='Sales'
        plt.bar(df['Month'], df['profit'], width=width, color='skyblue', label='pro
        plt.title('Sales and profit Trends by Months')
        plt.xlabel('Month')
        plt.ylabel('Amount ($)')
        plt.grid(True)
        plt.legend()
        plt.tight_layout(),

    elif plot_type == 'Pie chart':
        fig.set_size_inches(7,7)
        plt.pie(df['profit'], labels=df['Month'], autopct ='%1.2f%%', startangle =
        plt.title('profit by Month'),

    elif plot_type == 'Scatter Plot':
        plt.scatter(df['Sales'], df['profit'], color='green', s=100, edgecolors='re
        plt.title('Sales VS profit Scatter Plot')
        plt.xlabel('Sales')
        plt.ylabel('profit')
        plt.grid(True),

    elif plot_type == 'Histogram':
        plt.hist(df['Sales'], bins=5, color='purple', edgecolor='black')
        plt.title('Sales Distribution')
        plt.xlabel('Sales ($)')
        plt.ylabel('Frequency')

    elif plot_type == 'Box plot':
        plt.boxplot(df['profit'], vert=False, patch_artist=True, boxprops =dict(fac
    plt.title('Profit Distribution')
    plt.xlabel('Profit ($)')

    plt.tight_layout()
    return fig
```

```
# gradio ui

    demo = gr.Interface(
        fn = generate_plot,
        inputs = gr.Radio(['Line plot', 'stacked bar Chart', 'Pie chart', 'Scatter
        outputs = gr.plot(label="Sales data visualization"),
        title = 'sales & profit visual insight',
        description = 'Choose the type to visualize the data'
),

demo.launch()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[40], line 73
     63 # gradio ui
     65     demo = gr.Interface(
     66         fn = generate_plot,
     67         inputs = gr.Radio(['Line plot', 'stacked bar Chart', 'Pie chart', 'S
catter Plot', 'Histogram', 'Box plot']),
   (...)     70         description = 'Choose the type to visualize the data'
     71 ),
---> 73 demo.launch()

NameError: name 'demo' is not defined
```

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